Programming for Applications

Syllabus

Yu-You Liou

Fall 2026

Lecturer: Yu-You Liou	**Contact**: d10627008@ntu.edu.tw
Course Code: EIB-302-01-A1	**Website**: https://yyliou.github.io/pa
Meeting Time: Tuesday, Periods 8–9 (15:10–17:00)	**Location**: A207

Course Objectives

1. To enable students to install, configure, and operate the R environment confidently and to understand the structure of the R language.
2. To develop students' ability to write clear, correct, and efficient R programs using functions, objects, and environments.
3. To equip students with practical skills for importing, cleaning, transforming, visualizing, and statistically analyzing real-world data.
4. To integrate generative AI to enhance students' workflows in programming and compilation tasks.

Textbooks

1. Adler, J. *R in a Nutshell: A Desktop Quick Reference* (2nd ed.). O'Reilly Media. (Main)
2. Wickham, H., Navarro, D., & Pedersen, T. L. *ggplot2: Elegant Graphics for Data Analysis* (3rd ed.). https://ggplot2-book.org

Course Structure

1. **Lecture** (First hour): Foundational knowledge and concepts introduced each session.
2. **Lab** (Second hour): Hands-on in-class exercises using R to apply the concepts learned during the lecture.

Assessment Criteria

1. **In-Class Exercises** (36%): 12 exercises × 3% each, submitted at the end of each session.
2. **Mid-Term Exam** (32%) and **Final Exam** (32%).

Regulations

1. **Attendance**

Per Shih Chien University regulations, students who are absent for more than one-third of class sessions will receive a final grade of zero. Make-up attendance will not be permitted.

2. **Examination**

Students who do not show up for their scheduled examination will receive a grade of zero for that component.

3. **In-class exercise deadline**

The deadline for all in-class exercises is before midnight of the following week. Late submissions will not be accepted.

4. **Academic Integrity**

All submitted work must be the student's own. Copying from other students is strictly prohibited. AI detection tools are used to review submissions; any work found to be copied will receive a grade of zero.

5. **AI Policy**

Students may use AI tools (e.g., ChatGPT, GitHub Copilot) as learning aids, but must be able to explain and defend any code or analysis they submit. Uncritical submission of AI-generated work without understanding constitutes academic dishonesty.

Required Software

All software is free and open-source: **R** at https://cran.r-project.org, **RStudio Desktop** at https://posit.co/download/rstudio-desktop, and R packages `tidyverse`, `ggplot2`, `lattice` (installation instructions on the course R page).

Course Schedule

1. Week 1: Part I — R basics: installing R, the user interface, a first tutorial, and packages (Ch. 1–4).
2. Weeks 2–4: Part II — The R language: overview, syntax, objects, symbols and environments, functions, and object-oriented programming (Ch. 5–10).
3. Weeks 5–6: Part III — Working with data: saving, loading, editing, and preparing data (Ch. 11–12).
4. Week 7: Part IV — Data visualization with base graphics (Ch. 13).
5. Weeks 8–9: Mid-term exam.
6. Week 10: Part IV — Lattice graphics and `ggplot2` (Ch. 14–15).
7. Weeks 11–14: Part V — Statistics with R: analyzing data, probability distributions, statistical tests, power tests, regression, classification, machine learning, and time series (Ch. 16–23).
8. Weeks 15–16: Final exam.
9. Weeks 17–18: Flexible learning weeks; Part VI (Ch. 24–26) provided as self-study material.

Syllabus is subject to change. Any updates will be announced on the course website.